PEAAII AI Applications Hackathon 2026, Agentic AI

Roameo

Agents plan the trip.
You make the call.

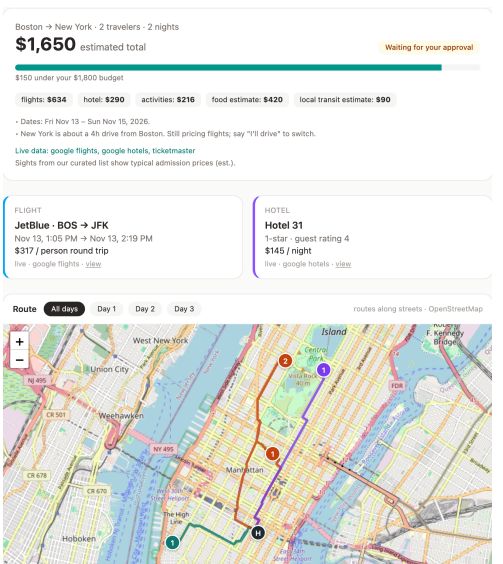
An AI trip planner for **students and budget travellers**. It does the research and the math, and **asks you** before it goes over your budget.

Zhen Lu **Noctis Zhang**

October 2026

Try it
roameo-l13h
.onrender.com

Code
github.com
/TinyZhen
/Travel_Agent



A plan in the app: total against the budget, flight, hotel and each day's route.

Alex has $600 and one sentence



Alex, a student

an illustration



$600

for a long weekend,
all in

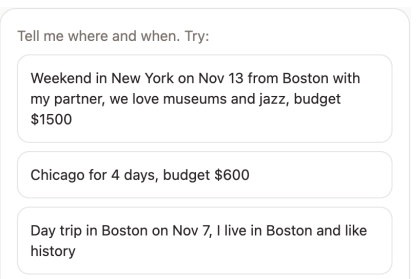
Everything has to add up



flights + hotel + sights, events + food + local transit

around the weather, the dates and what Alex likes, and again after every change

= total ≤ $600 ?



The request is the app's own example on its start screen.

A chat reply that adds the prices itself can say *it fits your budget* with nobody checking the sum.

Alex is an illustration. Booking.com's 2025 Global AI Sentiment Report, cited in our project report: 89% of travellers want AI help planning trips, 12% are comfortable with AI deciding alone.

One sentence becomes a checked plan, and Alex keeps the final say

1 Alex asks



Alex, a student (an illustration)

Chicago for 4 days, budget $600

$600

One sentence. The budget is the hard limit.

2 It asks back

Where are you leaving from?

What date do you want to leave?

From Boston, leaving Nov 20

It never guesses a city or a date. Missing or wrong? It asks.

3 Agents research

- 8 flights
- 18 hotels
- 59 events
- sights and markets
- rain and cold likely

Live flights, hotels and events, chosen and called by an AI agent.

4 The critic checks



plain code, not the AI

first plan

$1,235

after 5 repairs

$780 $180 over the $600

It shows the gap and does not decide alone.

5 Alex decides

raise the budget

shorten to 2 nights

drive instead of flying

The plan is rebuilt and checked again.

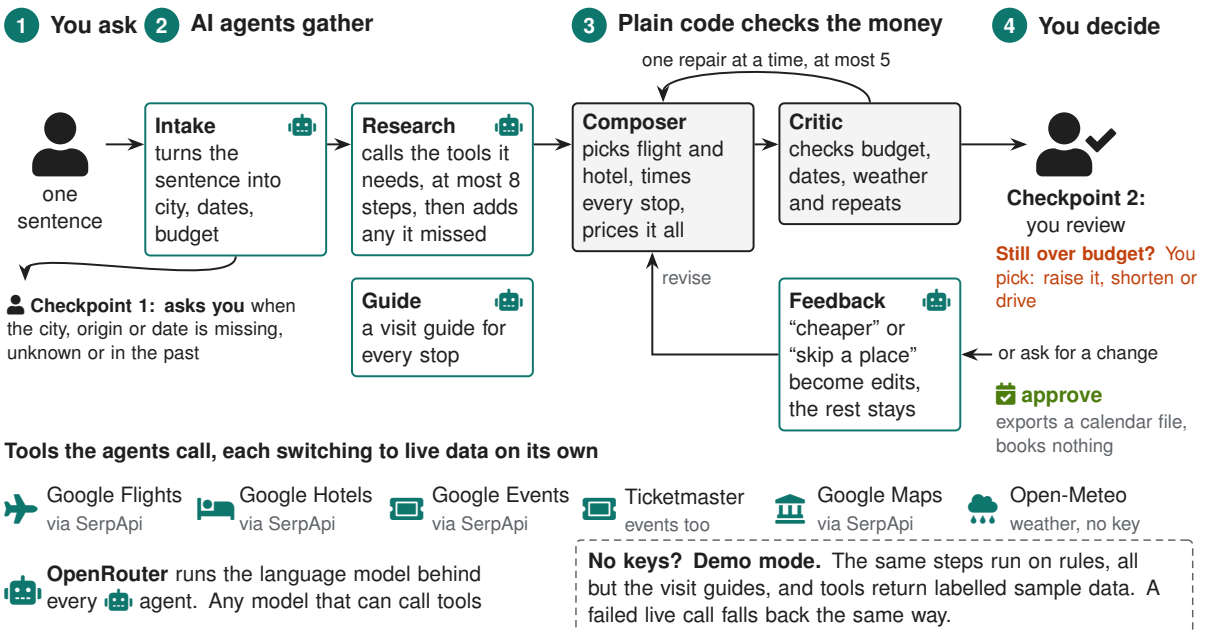
Approve: calendar file

Nothing is ever booked or paid for.

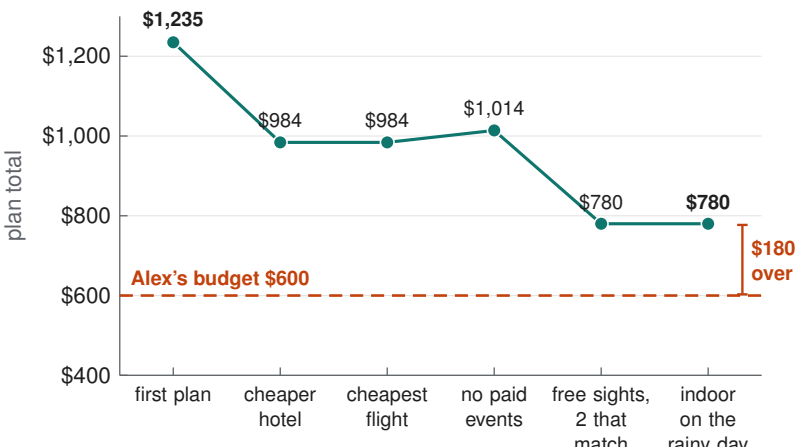
An **agent** is an AI helper with one job. The **critic** is plain code that checks the cost against the budget.

Every word and number above is from one real run of the app with live flights, hotels and events (the app's screenshot of this run is on the next slides).

AI AGENTS GATHER, PLAIN CODE CHECKS THE MONEY, YOU DECIDE TWICE



THE CRITIC RE-CHECKS EACH REPAIR, AND STOPS AFTER FIVE



Alex's run, read from the app's own log of what the agents did. Each point is one review by the critic, labelled with the repair just made.

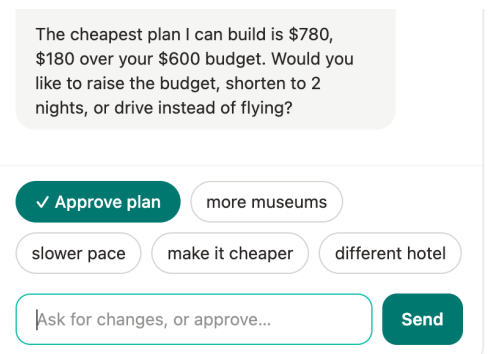
Each repair rebuilds the plan, so one step even went up. The limit is five repairs, then a human decides. In our tests a $300 budget gets the same honest answer: $778, which is $478 over.

$180 over

THEN IT STOPS AND ASKS ALEX, INSTEAD OF QUIETLY GOING OVER



The plan in Alex's run: the budget bar turns red, and the plan waits for approval.



The question in the chat, with buttons for a change or approval.

Alex's call

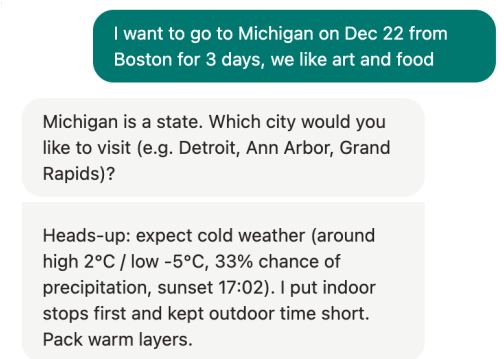
raise the budget | shorten to 2 nights | drive instead of flying

Alex picks one. The plan is **rebuilt and checked again**, then waits for Alex.

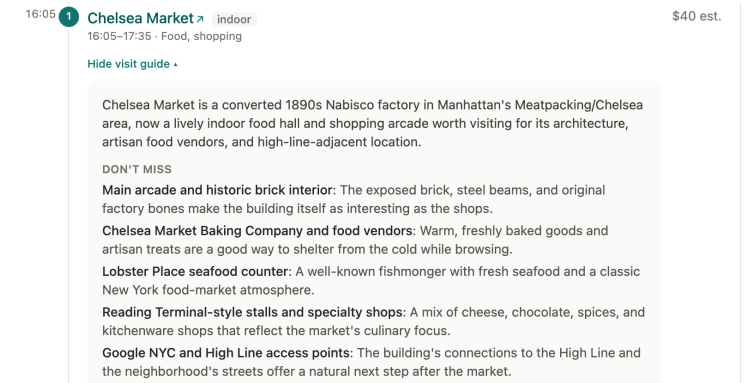
Approve only exports a calendar file. Roameo never books or pays for anything.

PLANS BEND TO FEEDBACK, WEATHER AND THE SEASON

Say it in plain words: cheaper | different hotel | skip a place | more museums | slower pace



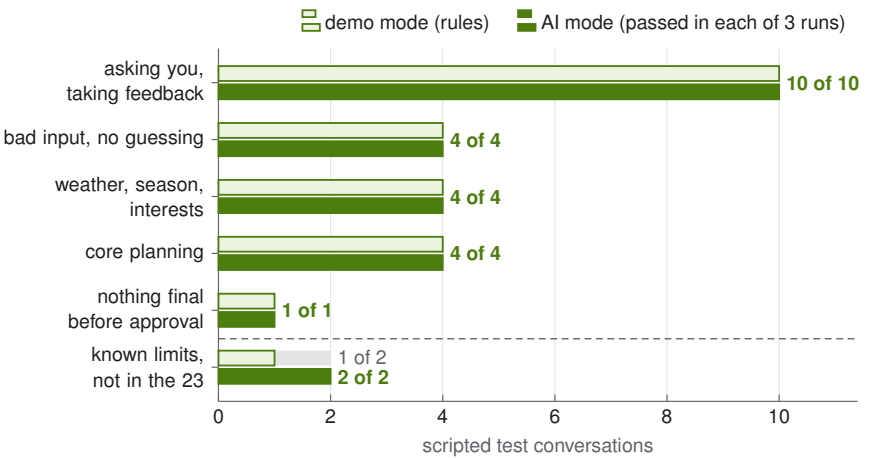
Michigan in December: it asks which city, then warns about the cold and puts indoor stops first.



A visit guide for one stop: what it is and what not to miss.

Each request is an **edit, not a restart**: the rest of the plan stays, and the reply lists what changed.

All 23 core tests pass in both modes, and in each of three AI runs



A **test** is a scripted conversation with checks on the result, such as *total within budget* or *no flight search for a road trip*: 51 checks, all passed.

We wrote the tests ourselves, on sample prices. Demo mode follows fixed rules, so only the 3 AI runs speak to consistency.

What it cannot do yet



Sample data for 7 cities

Without keys, only New York, Chicago, Boston and four more cities are covered.



Food and transit are estimates

Flat amounts per day, not real prices.



Events often unpriced

Shown as “price on site” and left out of the total.



A small test set

25 test conversations, written by us, on sample prices. 1 demo run, 3 AI runs. No user study yet.



Guides may be dated

Visit tips are written by the AI. Only museum exhibitions come from a web search.



No accounts

Trips live in memory. A restart clears them.

The tests do not judge how good a plan is, live prices, or the visit guides.

One URL to deploy, three minutes to try without keys

Try it in about 3 minutes

Install, start the server and the page, open it in a browser.

No keys needed. Demo mode runs the same steps on rules and labelled sample prices, all but the visit guides.

Deploy: one URL



A blueprint file sets it up. An access code and 60 requests per visitor per hour protect the search quota.

Go live one key at a time

- **OpenRouter**
  AI agents, any model
- **SerpApi**
  flights, hotels, events
- **Ticketmaster**
  events, any date
- **Google Places**
  sights, optional

Weather needs no key.

Built for **students and budget travellers** on short trips, and for small groups who split the cost.
The free Render plan sleeps after 15 idle minutes, so open it a minute before you need it.

Alex still makes the call on the $600

Roameo's agents do the **research** and the **arithmetic**. When the money does not work out, they **stop and ask**, instead of quietly going over.

raise the budget | shorten to 2 nights | drive instead

Try it
live demo

Code
GitHub

Thank you

Zhen Lu Noctis Zhang

Backup: the critic's repairs, in a fixed order

1. Swap outdoor stops for indoor ones on bad-weather days
2. Cheaper hotel, one tier at a time
3. Cheapest flight
4. Drop paid events
5. Free sights plus the 2 that best match your interests
6. Free sights only

The critic applies one repair, the composer rebuilds and prices the plan, and the critic checks again, at most five times. If the plan is still over budget, the person decides, so the last repair may never run. The weather swap runs first whenever a rainy day has outdoor stops. In Alex's run that happened only after the sights changed, so it came last. Budget repairs run only while the plan is over budget.

Escalation also covers no flight, no hotel, light days and cold weather. In the AI mode, a failed model call falls back to the rule path, so some AI-mode passes may use rules.

Backup: what the 25 test conversations check

Asking you and feedback (10)	Missing date, missing destination and missing origin are asked, never assumed. An impossible budget is escalated. Raising the budget makes the plan fit. “Cheaper” really lowers the price, “different hotel” changes it, a named place is skipped. “More museums” revises, “slower pace” lightens the days.
Bad input, no guessing (4)	Unknown city, a past date, vague feedback, a state instead of a city.
Weather, season, interests (4)	Rainy days start indoors, winter days keep outdoor time short, a river cruise is not scheduled out of season, cutting cost keeps a museum for a museum lover.
Core planning (4)	A complete request gives a priced plan within budget, driving skips flights and adds gas and parking, a local day trip has no flight or hotel, a family of four gets two rooms.
Safety (1)	Nothing is approved without an explicit approval.
Known limits (2)	A city outside the demo data is asked about. A complex date phrase: demo mode asks for a date instead, AI mode resolves it.

From the results files in the repository: demo mode with 1 run, AI mode with 3 runs per test.